

Rupankar Ganguly

Final-year B.Tech in Computer Science & Engineering

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SUMMARY

Final-year B.Tech Computer Science & Engineering student with hands-on software development experience across Java, Spring Boot, FastAPI, React, and SQL/NoSQL databases, plus foundational exposure to cloud platforms (Azure, AWS fundamentals), data analytics (Excel, Power BI), and applied AI/ML through GenAI-integrated projects. Built three end-to-end projects spanning backend service development, REST API design, and a network-diagnostics platform (NetSage AI) that combines LLM-based diagnosis with a mandatory human-review safety gate. CCNA-certified in networking fundamentals through a Cisco-AICTE virtual internship. Solved 150+ Data Structures & Algorithms problems on LeetCode, reached a max contest rating of 1480, and brings strong problem-solving, technical writing, and version-control habits to maintainable, cross-functional delivery. Graduating July 2027.

SKILLS

Languages & Frameworks: Java, Python, JavaScript, SQL, Spring Boot, FastAPI, React, HTML5, CSS3, Tailwind CSS, OOP, REST API Design, JWT Authentication

Databases: MySQL, PostgreSQL, MongoDB Atlas (NoSQL)

Data & Analytics: Microsoft Excel (formulas, pivot tables, data analysis), Power BI (dashboards & reports - foundational), data interpretation for decision support

Cloud Fundamentals: Working knowledge of Azure and AWS core concepts (compute, storage, networking basics); familiar with cloud-hosted service architecture from networking coursework

AI/ML & GenAI: Applied GenAI/LLM integration into production-style workflows (NetSage AI diagnosis engine, LLM-powered review analysis), foundational Machine Learning concepts, prompt-layer collaboration

Networking: Routing, Switching, VLANs, OSPF (CCNA 1-3 fundamentals), DHCP, DNS, ACL, NAT

Tools: Git/GitHub, Postman API Testing, Packet Tracer

PROFESSIONAL EXPERIENCE

Intern — Cisco Networking Academy *Remote | May 2026 - Jul 2026*

- Applied version control discipline and technical problem solving in Packet Tracer labs by modeling multi-router topologies and validating service behavior.
- Architected NetSage AI, an AI/GenAI-powered network-diagnostics platform, as the internship capstone in a 2-3 member team.
- NetSage AI diagnoses network faults across VLAN, DHCP, DNS, routing, ACL, NAT, and wireless issues from symptom reports and show-command evidence using an LLM-based diagnosis engine, improving service triage.
- Enforced a mandatory human-review responsibility before any AI-generated diagnosis is applied as a fix, adding a safety gate to the service workflow.
- Validated the dashboard against 33 test cases across 8 fault categories while a teammate implemented the AI diagnosis and prompt layer.

EDUCATION

Associate of Science in Computer Science & Engineering | CGPA - 7.37

Kalinga Institute of Industrial Technology — Kolkata *Jul 2027*

Coursework - Class XII: 65%, Class X: 75%

CERTIFICATIONS AND TRAINING

Cisco CCNA: Introduction to Networks • Cisco CCNA: Switching, Routing, and Wireless Essentials • Cisco CCNA: Enterprise Networking, Security, and Automation • Python Essentials 1 • Python Essentials 2 • Cisco + AICTE Virtual Internship Certificate of Completion

PROJECT EXPERIENCE

Real-Time Collaborative Whiteboard *(Live Demo)*

Engineered a vanilla JavaScript whiteboard application without external drawing libraries, emphasizing clean programming and maintainable code. Organized the codebase into 6 modular files totaling 790 lines of code to improve maintenance and versioning.

Implemented 5 core features: multi-colour pencil, adjustable-size eraser, sticky notes, image insertion, and board export for a better user experience.

Movie Recommendation System *(Live Demo)*

Developed the Spring Boot backend service for a full-stack movie recommendation platform with a focus on scalable microservices patterns. Published REST APIs consumed by a teammate's React.js frontend, supporting end-to-end feature delivery across the cross-functional team. Modeled MongoDB Atlas data structures to store user preferences and movie metadata, improving query performance for personalized recommendations.

Faculty Review System *(Live Demo)*

Built a full-stack faculty review platform in a 2-member team, owning the FastAPI backend service, frontend, and PostgreSQL database design. Designed REST APIs for review submission and retrieval with a focus on clean framework integration and service reliability. Added JWT-based authentication and email verification to prevent duplicate or fraudulent reviews and strengthen maintenance of platform integrity. Integrated a teammate's LLM-powered (GenAI) review-analysis feature into the core REST API, coordinating technical design decisions end to end.

LANGUAGES

Java, Python, JavaScript, SQL